Year 9 Science Test 1: mark scheme Core paper

Variation and inheritance · Adaptation and extinction · Cells and the microscope · Atoms and the periodic table · 40 minutes · 40 marks

Give the marks shown. Accept any correct equivalent wording. Where a part says "all four correct = 2 marks", do not award half marks.

1 Environmental variation (8 marks)

Part	Answer	Marks
(a)	temperature: abiotic; predation: biotic; amount of light: abiotic; infectious disease: biotic. All four correct = 2 marks; two or three correct = 1 mark.	2
(b)	environmental	1
(c)	height: continuous; blood group: discontinuous; hand span: continuous; tongue rolling: discontinuous. All four correct = 2 marks; two or three correct = 1 mark.	2
(d)	Yes (1). Leaf length is continuous variation / the lengths were put into groups that run on from each other, so there are no gaps (1).	2
(e)	A group of organisms that can breed together to produce offspring that can also breed (accept: organisms that can breed together to make fertile offspring).	1

2 Inherited variation (8 marks)

Part	Answer	Marks
(a)	nucleus (1), gametes (1), fertilisation (1), zygote (1).	4
(b)	natural eye colour (1) and blood group (1). If more than two boxes are ticked, take one mark off for each extra tick, to a minimum of 0.	2
(c)	Each gamete (sperm or egg) carries a different mixture of the parent's genetic information (1), so each child gets a different combination of genes from the two parents (1). Accept: they inherit different genes from each parent.	2

3 Adaptation and extinction (8 marks)

Part	Answer	Marks
(a)	A characteristic (feature) that helps an organism survive in its habitat.	1
(b)	The fur is camouflage: the cat is hard to see against the sand (1), so it can creep up on prey / is not seen by predators, and so catches more food or is not eaten (1).	2
(c)	Any two of: greys can eat unripe acorns, so they eat the food before the reds do (compete better for food); greys store more body fat, so more survive the winter; greys carry squirrel pox, a disease that kills reds.	2
(d)	nature reserve: an area where the habitat and the species living in it are protected (1); breeding programme: endangered animals are bred in captivity (1); gene bank: seeds, gametes or cells stored at a low temperature (1).	3

4 Cells and the microscope (8 marks)

Part	Answer	Marks
(a)	A cell wall (1), B nucleus (1), C chloroplast (1), D vacuole (1).	4
(b)	cell membrane	1
(c)	x40 (10 x 4)	1
(d)	Any two of: use a sharp pencil; make the drawing large; use clear single lines with no shading or colouring; draw only what you see; draw label lines with a ruler so they do not cross and end on the part; add a title with the magnification.	2

5 Atoms and the periodic table (8 marks)

Part	Answer	Marks
(a)	neutron: 0 (1), in the nucleus (1). electron: -1 (1), in shells around the nucleus (1).	4
(b)	(i) 11 protons (1). (ii) 23 - 11 = 12 neutrons (1).	2
(c)	2,8,1 (the missing number is 1).	1
(d)	2,8,2	1

Total: 40 marks